Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #4

Posted: Monday, Feb 23 **Due: Tuesday, Mar 3, 11:59 pm**

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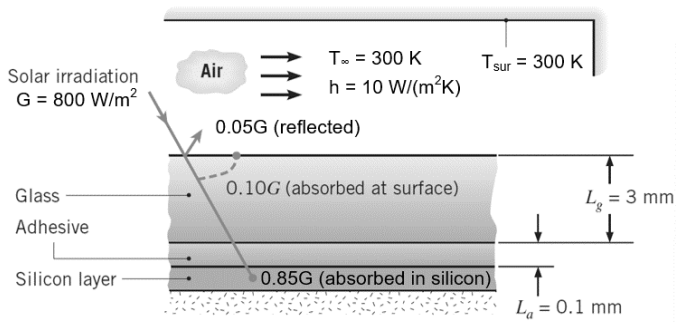
1. (40 pt) A photovoltaic (PV) panel consists of (top to bottom) a 3-mm-thick ceria-doped glass (k_g = 1.5 W/m·K), a 0.1-mm-thick optical grade adhesive (k_a = 1 W/m·K), and a very thin layer of silicon within which solar energy is converted to electricity. The solar-to-electricity conversion efficiency within the silicon layer η decreases with increasing silicon temperature T_{Si}, as described by the expression $\eta = a - bT_{Si}$, where a = 0.553 and b = 0.001 K^{-1}. The temperature T_{Si} is expressed in Kelvins. The backside of the silicon layer is thermally insulated. Contact thermal resistance between the adhesive and silicon layer is $R_c = 10^{-4}$ m^2·K/W. There is no contact resistance between the glass and the adhesive.



In a sunny day, there is G = 800 W/m^2 solar irradiation reaching the PV panel. 5% of solar irradiation is reflected from the top surface of the glass, 10% of solar irradiation is absorbed at the top surface of the glass, and 85% of solar irradiation is transmitted to and absorbed within the silicon layer. Part of the solar irradiation absorbed in the silicon is converted to thermal energy and the remainder is converted to electricity. The PV panel is exposed to ambient air with temperature of T_∞ = 300 K and convective heat transfer coefficient of h = 10 W/m^2·K. The surrounding temperature is $T_{sur} = T_\infty$ = 300 K and the effective radiative heat transfer coefficient is h_{rad} = 5 W/m^2·K. Hint: you do not need to use the Stefan-Boltzmann law to calculate the radiative heat loss.

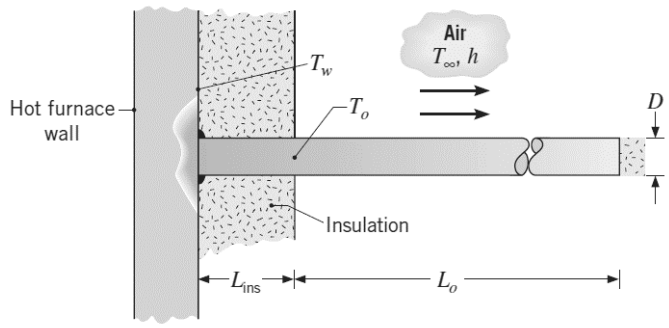
(a) Show the thermal resistance network that can describe heat transfer across the PV panel. Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the

temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network.

(b) Calculate the temperatures of the top surface of the glass $T_{g,top}$ and the silicon layer T_{Si}.

(c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

2. (30 pt) A rod of diameter D = 10 mm and thermal conductivity k = 80 W/m·K protrudes normally from a furnace wall that is at T_w = 200 °C and is covered by insulation of thickness L_{ins} = 200 mm. The rod is welded to the furnace wall and is used as a hanger for supporting instrumentation cables. The rod has an exposed length L_o and its tip is well insulated. To avoid damaging the cables, the temperature of the rod at its exposed surface, T_o, must be maintained below a specified operating limit of T_{max} = 100 °C. The ambient air temperature is T_∞ = 20 °C and convective heat transfer coefficient is h = 20 W/m^2·K.



(a) Derive an expression for the exposed surface temperature T_o as a function of the prescribed thermal geometrical parameters.

(b) If exposed length L_o is 200 mm, calculate the corresponding exposed surface temperature T_o. Does the exposed length meet the specified operating limit?

(c) Calculate the minimum exposed length L_o that ensures T_o below the specified operating limit.

3. (30 pt) A square pin fin of uniform cross-section area is fabricated of copper (k = 400 W/m·K). The fin width is s = 10 mm and the fin is exposed to air flow with convective heat transfer coefficient h = 100 W/m^2·K. It is known that the fin efficiency is η_f = 0.6. The fin has a convection tip.

(a) Calculate the fin length L.

(b) Calculate the thermal resistance of the fin R_f and the fin effectiveness ε_f.

1) **Known:** $G = 800 \frac{W}{m^2}$, Reflected at top surface 5%

Absorbed at top of glass 10% -> $q''_{g,abs} = 0.1G = 80 \frac{W}{m^2}$

Transmitted and absorbed in silicon 85% -> $q''_{si,abs} = 0.85G = 680 \frac{W}{m^2}$

$L_g = 3\times10^{-3} m$, $k_g = 1.5 \frac{W}{m\cdot K}$

$L_a = 1\times10^{-4} m$, $k_a = 1 \frac{W}{m\cdot K}$

$R_c = 10^{-4} \frac{m^2\cdot K}{W}$, $T_\infty = 300 K = T_{sur}$, $h = 10 \frac{W}{m^2\cdot K}$, $h_{rad} = 5 \frac{W}{m^2\cdot K}$

Efficiency $\eta = a - b T_{si}$, $a = 0.553$, $b = 0.001 K^{-1}$

find: a) Show thermal resistance network, list assumptions, and calculate each resistance.

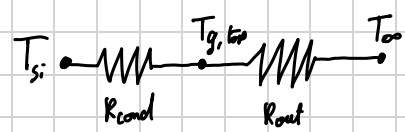
b) Calculate $T_{g,top}$ and T_{si}

c) Calculate the efficiency η and electric power produced by the PV panel.

Assumption

1) steady state
2) 1-D heat flow and uniform flow
3) k, h, h_{rad} are constant
4) below the silicon is perfectly insulated.

Analysis



a) $q = -kA \frac{dT}{dx} = -kA \times \frac{(T_2 - T_1)}{L} \Rightarrow T_1 - T_2 = q\left(\frac{L}{kA}\right)$ ($\frac{dT}{dx} = \frac{T_2 - T_1}{L}$)

we know from ohm's law -> $\Delta T = qR$ so $R = \frac{L}{kA}$

$q'' = \frac{q}{A}$ so $R'' = \frac{L}{k}$ thus $R_g = \frac{L_g}{k_g} = \frac{3\times10^{-3}}{1.5} = 2\times10^{-3} \frac{m^2\cdot K}{W}$

$R_a = \frac{L_a}{k_a} = \frac{1\times10^{-4}}{1} = 10^{-4} \frac{m^2\cdot K}{W}$

$\boxed{R_{cond} = R_g + R_a + R_c = 2\times10^{-3} + 10^{-4} + 10^{-4} = 2.2\times10^{-3} \frac{m^2\cdot K}{W}}$ ← between T_{si} and $T_{g,top}$

$h_{out} = h + h_{rad} = 10 + 5 = 15 \frac{W}{m^2\cdot K}$

$\boxed{R_{out} = \frac{1}{h_{out}} = 6.667\times10^{-2} \frac{m^2\cdot K}{W}}$ ← outside at $T_{g,top}$

b) $q'' = q''_{si,th} = (1-\eta) q''_{si,abs} = \left[(1-a) + bT_{si}\right] \times 680 \frac{W}{m^2}$

q'' across R_{cond} is $q'' = \frac{(T_{si} - T_{g,top})}{R_{cond}}$ and across R_{out} $q'' = \frac{(T_{g,top} - T_\infty)}{R_{out}} - q''_{g,abs}$

$\Rightarrow q'' = \frac{T_{g,top} - 300K}{2.2\times10^{-3} \frac{m^2\cdot K}{W}} - 80 \frac{W}{m^2}$

$$q'' = \frac{T_{si} - T_g}{2.2\times10^{-3}\,\frac{m^2\cdot K}{W}} \quad (1)$$

to T_g -> convection + rad $\Rightarrow$ $q'' + 80 = h_{conv}(T_g - T_\infty) \Rightarrow q'' = 15(T_g - 300) - 80\frac{W}{m^2} = 15T_g - 4580 \quad (2)$

Efficiency $q'' = (1-\eta)\times 680\,\frac{W}{m^2} = (0.447 + 0.001T_{si})\times 680\,\frac{W}{m^2} = 303.96 + 0.68\,T_{si}\,\frac{W}{m^2}$

$\hookrightarrow 1-(a-bT_{si}) = 1-(0.553-0.001T_{si}) = 0.447 + 0.001\,T_{si}$

$(2) = (3) \Rightarrow 15T_g - 4580 = 303.96 + 0.68T_{si}$

$\Rightarrow 15T_g = 0.68T_{si} + 4883.96$

$\Rightarrow T_g = 0.04533T_{si} + 325.597 \quad (4)$

now we will set $(1) = (3) \Rightarrow \frac{T_{si} - T_g}{2.2\times10^{-3}\,\frac{m^2\cdot K}{W}} = 303.96 + 0.68T_{si}$

also $T_{si} - T_g = T_{si} - 0.04533T_{si} - 325.597$
$= 0.95467T_{si} - 325.597$

$0.95467T_{si} - 325.597 = 0.668712 + 0.001496\,T_{si}$

$\Rightarrow \boxed{T_{si} = \frac{326.266}{0.95317} = 342.30\,K}$

so $\boxed{T_g = 0.04533T_{si} + 325.597 = 341.12\,K}$, $q'' = 15(341.12K - 300K) - 80\,\frac{W}{m^2} = 536.8\,\frac{W}{m^2}$

c) $\eta = a - bT_{si} = 0.553 - 0.001K^{-1}\times 342.3K = 0.2107$ or 21.07%

$q''_{elec} = \eta \times q''_{si,abs} = 0.2107\times 680\,\frac{W}{m^2} = 143.276\,\frac{W}{m^2}$

2) Known: rod diameter $D = 10\times10^{-3}\,m$
thermal cond $k = 80\,\frac{W}{m^2\cdot K}$
wall temp at left end $T_w = 200°C$, insulation thickness $L_{ins} = 200\times10^{-3}\,m$
ambiant air $T_\infty = 20°C$, $h = 20\,\frac{W}{m^2\cdot K}$, $T_{max} = 100°C$
L_0 exposed to T_∞
Fin tip is insulated

find a) derive an expression for the exposed surface temp T_0 as a function of the given parameters
b) Compute T_0 for $L_0 = 0.2m$ and check if T_0 exceed $T_{max} = 100°C$
c) Calculate the minimum L_0 length so that $T_0 \leq T_{max}$

Assume
steady state
1-D heat flow along rod axis
No heat loss along the rod inside the insulation
Exposed portion behaves like a straight fin with constant k and h
Tip is adiabatic

Analysis a) $A = \frac{\pi D^2}{4}$, $P = \pi D$

$q = kA \frac{(T_w - T_0)}{L_{ins}}$ } -> fourier's law

with adiabatic tip $q = mkA(T_0 - T_\infty)\tanh(mL_0)$, $m = \sqrt{\frac{hP}{kA}} = \sqrt{\frac{h\cancel{\pi D}}{k\frac{\cancel{\pi D^2}}{4}}} = \sqrt{\frac{4h}{kD}} = \sqrt{\frac{4 \times 20 \frac{\cancel{W}}{m^2 \cancel{K}}}{80 \frac{\cancel{W}}{\cancel{m}\cancel{K}} \times 10 \times 10^{-3} \cancel{m}}} = 10 \frac{1}{m}$

so $\cancel{kA}\frac{(T_w - T_0)}{L_{ins}} = m\cancel{kA}(T_0 - T_\infty)\tanh(mL_0)$

$\Rightarrow L_{ins} \cdot m \cdot \tanh(mL_0) = B$

$\Rightarrow T_w - T_0 = BT_0 - BT_\infty$

$\Rightarrow T_0 = -BT_0 + BT_\infty + T_w$

$$\Rightarrow T_0 = \frac{BT_\infty + T_w}{(1+B)} = \frac{0.2m \times 10\frac{1}{m} \times \tanh\left(10\frac{1}{m} \times L_0\right) 293.15K + 473.15K}{1 + 0.2m \times 10\frac{1}{m} \times \tanh\left(10\frac{1}{m} \times L_0\right)}$$

$$\boxed{= \frac{2 \times \tanh\left(10\frac{1}{m} \times L_0\right) 293.15K + 473.15K}{1 + 2 \times \tanh\left(10\frac{1}{m} \times L_0\right)}}$$

b) $L_0 = 0.2m$,

$$\text{so } T_0 = \frac{2 \times \tanh\left(10\frac{1}{m} \times 0.2m\right) 293.15K + 473.15K}{1 + 2 \times \tanh\left(10\frac{1}{m} \times 0.2m\right)} = \boxed{354.62K \text{ or } 81.47°C}$$

c) We want T_0 at max $T_0 = T_{max} = 100°C = 373.15K$

$$T_0 = \frac{2 \times \tanh\left(10\frac{1}{m} \times L_0\right) 293.15K + 473.15K}{1 + 2 \times \tanh\left(10\frac{1}{m} \times L_0\right)}, \quad B = 2\tanh\left(10\frac{1}{m} \times L_0\right)$$

$$T_0 = \frac{BT_\infty + T_w}{1+B} \Rightarrow T_0(1+B) = BT_\infty + T_w \Rightarrow BT_0 - BT_\infty = T_w - T_0 \Rightarrow B = \frac{T_w - T_0}{T_0 - T_\infty} = \frac{200 - 100}{100 - 20} = \frac{100}{80} = 1.25$$

$\Rightarrow 2\tanh\left(10\frac{1}{m} \times L_0\right) = 1.25$

$\Rightarrow \tanh\left(10\frac{1}{m} \times L_0\right) = 0.625$

$\Rightarrow 10\frac{1}{m} \times L_0 = \tanh^{-1}(0.625)$

$\Rightarrow L_0 = \tanh^{-1}(0.625) \times \frac{1}{10} m$

$\Rightarrow L_0 = 0.0733m$

$\boxed{L_{0_{min}} = 73.3mm}$

3) **Known** $k_c = 400 \frac{W}{m \cdot K}$, $h = 100 \frac{W}{m^2 \cdot K}$

Square fin width $s = 0.01\,m$

Convection coef: $\eta_f = 0.6$

tip condition $\Rightarrow$ convection tip

find a) calculate fin L

b) calculate fin thermal resistance R_f and effectiveness ε_f

Assumptions

Steady state

1-D heat transfer

k and h are constant

Convection tip handled using the "corrected length" $L_c = L + \frac{A_c}{P}$

Analysis

a) $A_c = s^2 = (1 \times 10^{-2} m)^2 = 1 \times 10^{-4} m^2$, $m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{100 \frac{W}{m^2 \cdot K} \times 4 \times 10^{-2} m}{400 \frac{W}{m \cdot K} \times 1 \times 10^{-4} m^2}} = 10 \frac{1}{m}$

$P = 4s = 4 \times 10^{-2} m$

$\frac{A_c}{P} = 2.5 \times 10^{-3} m$

$L_c = L + 2.5 \times 10^{-3} m$

$\eta_f = 0.6 = \frac{\tanh(mL_c)}{mL_c}$, $x = mL_c$

$\hookrightarrow$ solve for $x \rightarrow 0.6 = \frac{\tanh(x)}{x}$

$\Rightarrow 0.6x = \tanh(x)$

$\Rightarrow 0.6x - \tanh(x) = 0$

$\Rightarrow x = \pm 1.5122$

x must be > 0 so $x = 1.5122$

Thus $mL_c = 1.5122$

$L_c = 0.15122\,m$

so $L = L_c - \frac{A_c}{P} = 0.15122\,m - 2.5 \times 10^{-3} m = 0.14872\,m$ so $\boxed{L \approx 149\,mm}$

b) fin surface area A_f

$A_f \approx PL_c$ so $A_f = 4 \times 10^{-2} m \times 0.15122\,m = 6.0488 \times 10^{-3} m^2$

$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{0.6 \times 100 \frac{W}{m^2 \cdot K} \times 6.0488 \times 10^{-3} m^2} = \boxed{2.755 \frac{K}{W}}$

$\varepsilon_f = \frac{q_f}{q_{nf}} = \frac{\eta_f \cancel{h} A_f \cancel{(T_b - T_\infty)}}{\cancel{h} A_c \cancel{(T_b - T_\infty)}} = \frac{\eta_f A_f}{A_c} = \frac{0.6 \times 6.0488 \times 10^{-3} m^2}{1 \times 10^{-4} m^2} = \boxed{36.2928}$